

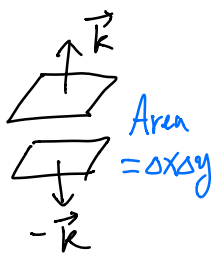
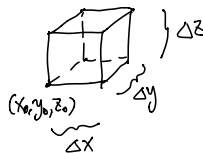
MAT1002 Lecture 27,

Outline

- Divergence and the divergence theorem
- Summary of "differentiation operators" and "big theorems"

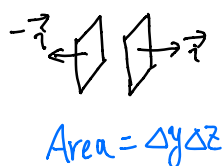
Divergence

Seeing $\text{div}(\vec{F})$ as (outward) flux density: Consider the flux of $\vec{F}$ across a very tiny cube.



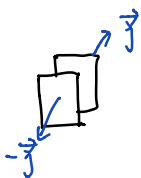
Flux 1

$$\begin{aligned} &\approx \vec{F}(x_0, y_0, z_0 + \Delta z) \cdot \vec{k} \Delta x \Delta y - \vec{F}(x_0, y_0, z_0) \cdot \vec{k} \Delta x \Delta y \\ &= \frac{P(x_0, y_0, z_0 + \Delta z) - P(x_0, y_0, z_0)}{\Delta z} \Delta x \Delta y \Delta z \quad \Delta V \\ &\quad \hookrightarrow \frac{\partial P}{\partial z}(x_0, y_0, z_0) \text{ as } \Delta z \rightarrow 0 \end{aligned}$$



Flux 2

$$\begin{aligned} &\approx \vec{F}(x_0 + \Delta x, y_0, z_0) \cdot \vec{i} \Delta y \Delta z - \vec{F}(x_0, y_0, z_0) \cdot \vec{i} \Delta y \Delta z \\ &= \frac{M(x_0 + \Delta x, y_0, z_0) - M(x_0, y_0, z_0)}{\Delta x} \Delta x \Delta y \Delta z \\ &\quad \hookrightarrow \frac{\partial M}{\partial x}(x_0, y_0, z_0) \text{ as } \Delta x \rightarrow 0 \end{aligned}$$



Flux 3

$$\begin{aligned} &\approx \vec{F}(x_0, y_0 + \Delta y, z_0) \cdot \vec{j} \Delta x \Delta z - \vec{F}(x_0, y_0, z_0) \cdot \vec{j} \Delta x \Delta z \\ &= \frac{N(x_0, y_0 + \Delta y, z_0) - N(x_0, y_0, z_0)}{\Delta y} \Delta x \Delta y \Delta z \\ &\quad \hookrightarrow \frac{\partial N}{\partial y}(x_0, y_0, z_0) \text{ as } \Delta y \rightarrow 0 \end{aligned}$$

Hence, when cube is tiny, flux across cube

$$\approx \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z} \right) \Delta V = \text{div}(\vec{F}) \Delta V$$

Note: this is the big idea, not a formal proof.

Divergence Theorem

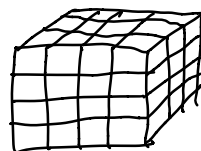
Theorem (Divergence Theorem)

Let $\mathbf{F} = \langle M, N, P \rangle$ be a vector field whose components have continuous partial derivatives. If E is a ^{closed} bounded solid having S as its boundary, where S is a closed, piecewise smooth surface, oriented **outward** (i.e. its unit normals point out from E), then

$$\underbrace{\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma}_{\text{Total flux}} = \iiint_E \underbrace{\text{div } \mathbf{F} \, dV}_{\text{"Microscopic flux" at a point}}.$$

Total flux

"Microscopic flux"
at a point



When summing microscopic flux, adjacent faces get cancelled.

Remark

Here, roughly speaking, a closed surface is a surface that divides the space into two regions, "inside" (which is bounded) and "outside" (which is unbounded).

e.g.

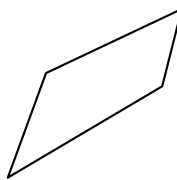


$$x^2 + y^2 + z^2 = a^2$$



cubic shell

closed



not closed

e.g.

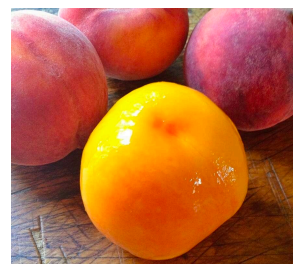
$x^2 + y^2 + z^2 \leq a^2$: closed bdd solid (closed ball B_a).

$x^2 + y^2 + z^2 = a^2$: closed surface that is the boundary of B_a .

e.g.

Whole peach : closed bdd solid.

peach skin : closed surface that is the boundary of the peach.



e.g.1 Find the flux of $\vec{F} = \langle z, y, x \rangle$ across the sphere $x^2 + y^2 + z^2 = 1$.
 if no direction is specified, flux is outward

• Last time, we found the flux to be $\frac{4\pi}{3}$ by computing a surface integral (lots of steps).

• If we use divergence thm:

$$\iint_S \vec{F} \cdot \vec{n} \, d\sigma = \iiint_B \operatorname{div} \vec{F} \, dV = \iiint_B 1 \, dV = |B| = \frac{4}{3}\pi 1^3 = \frac{4}{3}\pi.$$

$\nearrow$
 $x^2 + y^2 + z^2 \leq 1$

e.g.2 Find the flux of $\vec{F} = \langle x^2, 4xyz, ze^x \rangle$ across the boundary of the box $0 \leq x \leq 3$, $0 \leq y \leq 2$, $0 \leq z \leq 1$.

• Computing using surface integrals is tedious (six faces).

• If we use divergence thm:

$$\begin{aligned} \iint_S \vec{F} \cdot \vec{n} \, d\sigma &= \iiint_E \operatorname{div} \vec{F} \, dV = \iiint_E (2x + 4xz + e^x) \, dV \\ &= \int_0^3 \int_0^2 \int_0^1 (2x + 4xz + e^x) \, dz \, dy \, dx \\ &= \left(\int_0^2 dy \right) \left(\int_0^3 \int_0^1 (2x + 4xz + e^x) \, dz \, dx \right) \\ &= 2 \int_0^3 \left[(2x + e^x)z + 2xz^2 \right]_{z=0}^1 dx = 2 \int_0^3 (2x + e^x + 2x) dx \\ &= 2 (2x^2 + e^x) \Big|_0^3 = 2(18 + e^3 - 1) = 2(17 + e^3). \end{aligned}$$

$\nearrow$
 box
 no y

More General Solids

The divergence thm can be used on solids that can be decomposed into finitely many bounded solids that have a closed surface as the boundary.

$$(0 < b < a)$$

e.g. $E: b^2 \leq x^2 + y^2 + z^2 \leq a^2$ (B_a , with B_b removed)



$$\begin{aligned} & \text{Flux across boundary of } E \\ &= \text{flux across " " } E_1 + \text{flux across " " } E_2 \\ &= \iiint_{E_1} \operatorname{div} \vec{F} dV + \iiint_{E_2} \operatorname{div} \vec{F} dV \quad (\text{div thm, base form}) \\ &= \iiint_E \operatorname{div} \vec{F} dV \end{aligned}$$

e.g. Find the flux of $\vec{F} = \frac{1}{(x^2 + y^2 + z^2)^{3/2}} \langle x, y, z \rangle$ across the boundary S of E above.

Sol. Let $\rho := \rho(x, y, z) = \sqrt{x^2 + y^2 + z^2}$. Then

$$\frac{\partial \rho}{\partial x} = \frac{\rho^3 - x \cdot \frac{3}{2} \rho \cdot 2x}{\rho^6} = \frac{\rho^2 - 3x^2}{\rho^5}.$$

Similarly, $\frac{\partial v}{\partial y} = \frac{\rho^2 - 3y^2}{\rho^5}$, $\frac{\partial p}{\partial z} = \frac{\rho^2 - 3z^2}{\rho^5}$

Now flux = $\iint_S \vec{F} \cdot \vec{n} d\sigma = \iiint_E \text{div} \vec{F} dV$
 $= \iiint_E (M_x + N_y + P_z) dV = \iiint_E \frac{3\rho^2 - 3\rho^2}{\rho^5} dV = 0$

Let S_a be the sphere of radius a . Then the boundary S of E is $S_a \cup S_b$, with outward normal of E being inward normal of S_b (by itself). Hence

$$\iint_S \vec{F} \cdot \vec{n} d\sigma = \iint_{S_a} \vec{F} \cdot \vec{n} d\sigma - \iint_{S_b} \vec{F} \cdot \vec{n} d\sigma.$$

For the field $\vec{F} = \frac{1}{(x^2+y^2+z^2)^{3/2}} \langle x, y, z \rangle$, $\iint_S \vec{F} \cdot \vec{n} d\sigma = 0$,

so $\iint_{S_a} \vec{F} \cdot \vec{n} d\sigma = \iint_{S_b} \vec{F} \cdot \vec{n} d\sigma$ ($\forall a, b > 0$): this field causes the same flux across all spheres enclosing the origin.

Summary of "Differentiation Operators"

Purple = boundary of Green

① $f(x) \xrightarrow{\frac{d}{dx}} f'(x)$
 scalar scalar
 (real-valued)

Fundamental Thm of Calculus:

$$\int_{[a,b]} f(x) dx = \int_a^b f'(x) dx = f(b) - f(a)$$



$$\textcircled{2} \quad \underset{\text{Scalar}}{f(x,y,z)} \xrightarrow{\text{grad}} \underset{\text{vector}}{\nabla f(x,y,z)}$$

Fundamental Thm of Line Integrals:

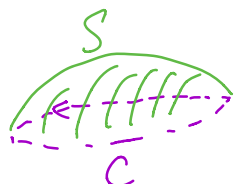
$$\int_C \nabla f \cdot d\vec{r} = f(B) - f(A)$$



$$\textcircled{3} \quad \underset{\text{vector}}{\vec{F}(x,y,z)} \xrightarrow{\text{curl}} \underset{\text{vector}}{\text{curl } \vec{F}} = \nabla \times \vec{F}$$

Stokes' Thm (3D)

$$\iint_S \text{curl } \vec{F} \cdot \vec{n} \, d\sigma = \oint_C \vec{F} \cdot d\vec{r}$$



Green's Thm (Circulation) (xy-plane)

$$\iint_R \text{curl } \vec{F} \cdot \vec{k} \, dA = \oint_C \vec{F} \cdot d\vec{r}$$



$$\textcircled{4} \quad \underset{\text{vector}}{\vec{F}} \xrightarrow{\text{div}} \underset{\text{scalar}}{\text{div } \vec{F}} = \nabla \cdot \vec{F}$$

Divergence Thm (3D)

$$\iiint_E \text{div } \vec{F} \, dV = \iint_S \vec{F} \cdot \vec{n} \, d\sigma$$



Green's Thm (Flux) (2D)

$$\iint_R \text{div } \vec{F} \, dA = \oint_C \vec{F} \cdot \vec{n} \, ds$$



We finish by proving the following thm (discussed in Lecture 22)

Theorem

Let $\mathbf{F}$ be a vector field whose components are continuous on an open connected region D . Then $\mathbf{F}$ is *conservative* on D if and only if $\mathbf{F}$ is a *gradient field* (that is, $\mathbf{F} = \nabla f$ for some f defined on D).

- Will show for 2D fields.
- " $\Leftarrow$ " : done, by the Fundamental Thm of Line Integrals.
- " $\Rightarrow$ " : Will show this. This proof is optional, meaning that you should understand it, but you do not need to memorize it.

Also, proofs for Stokes' thm and the divergence thm can be found in Stewart (the reference book in the course outline).